

What a passive power meter can certify about AI training

A claim ladder for adversarial verification

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THE POLICY QUESTION

A regulator sees one scalar trace



Inside the facility

Training, inference,
control loops,
cooling, and
unrelated tenants



$y(t)$

Outside the prover

A time-resolved
power trace



Governance decision

Did a restricted
training run occur?

Question: what conclusions does the trace actually license?

Power is trusted as a channel—not automatically as evidence

Why it is attractive

- Measured outside the prover's software stack
- Continuous and comparatively cheap
- Physical work must consume energy

Why it is dangerous

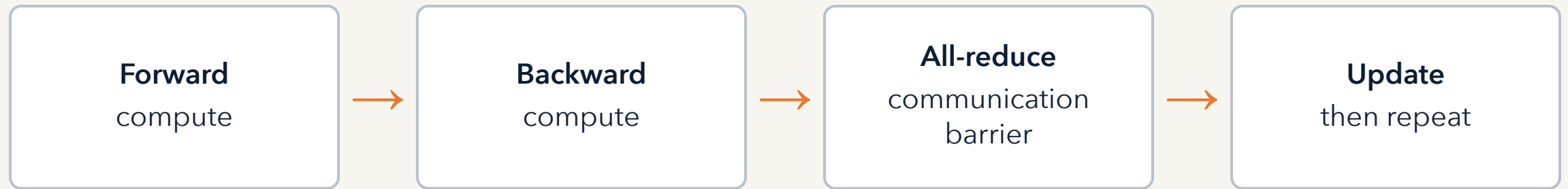
- Thousands of devices collapse into one number
- The meter filters, integrates, and samples
- Different programs can create the same physical schedule

observed trace = workload physics + nuisance loads + meter transfer function

The meter may be trustworthy while the interpretation is not.

THE PHYSICAL HYPOTHESIS

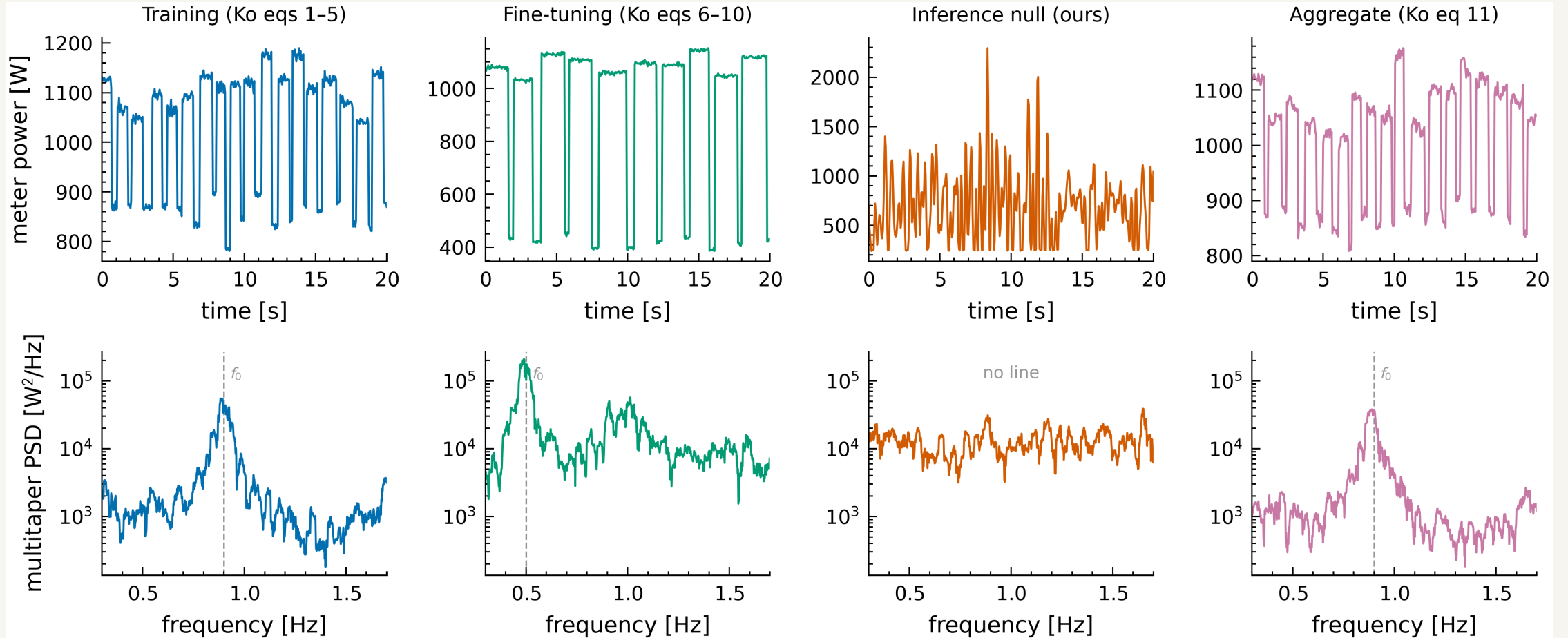
Efficient synchronous training should leave a rhythm



Repeated compute-communication phases can create a **cadence line** whose frequency wanders as iteration time changes.

The target is not a perfectly fixed frequency. It is order-coherent variation that remains repeatable after following the cadence.

The scenario model makes that hypothesis visible



SYNTHETIC SCENARIO Literature-parameterized workload and off-chip observation model. Time-domain traces above; spectra below.

THE FIRST TRAP

A tracker can lock onto a real line—and still track the wrong thing

Target A

True iteration cadence on one measured A100
 $\approx 0.4\%$ of board power

Confounder B

A power-management limit cycle under time-varying load
 $\approx 10\times$ stronger than A

Viterbi result

Correctly follows the strongest plausible path
But that path is B

Tracking solves frequency wander. It does not solve causal attribution.

THE DEEPER TRAP

Even the right cadence does not prove training

Genuine training

forward → gradients → update
y(t)

Semantic decoy

forward → gradients → discard
y(t)

If two behaviours induce the same trace distribution, every passive detector has the same decision distribution.

No better algorithm can recover information that never reaches the meter.

THE ORGANIZING IDEA

Every stronger claim must buy more information

1 Structural evidence

INFORMATION
One trace

CANNOT EXCLUDE
Confounders, shaping, superposition

2 Conditional classification

INFORMATION
Trace + stated null population

CANNOT EXCLUDE
Domain shift, semantic decoys

3 Active authentication

INFORMATION
Unpredictable challenge

CANNOT EXCLUDE
Challenge-aware dummy work

4 Work verification

INFORMATION
Challenge + bound transcript

CANNOT EXCLUDE
Unbound co-located responder

RUNG 1 · STRUCTURAL EVIDENCE

Test for order-coherent structure—not for “training”



Evidence for

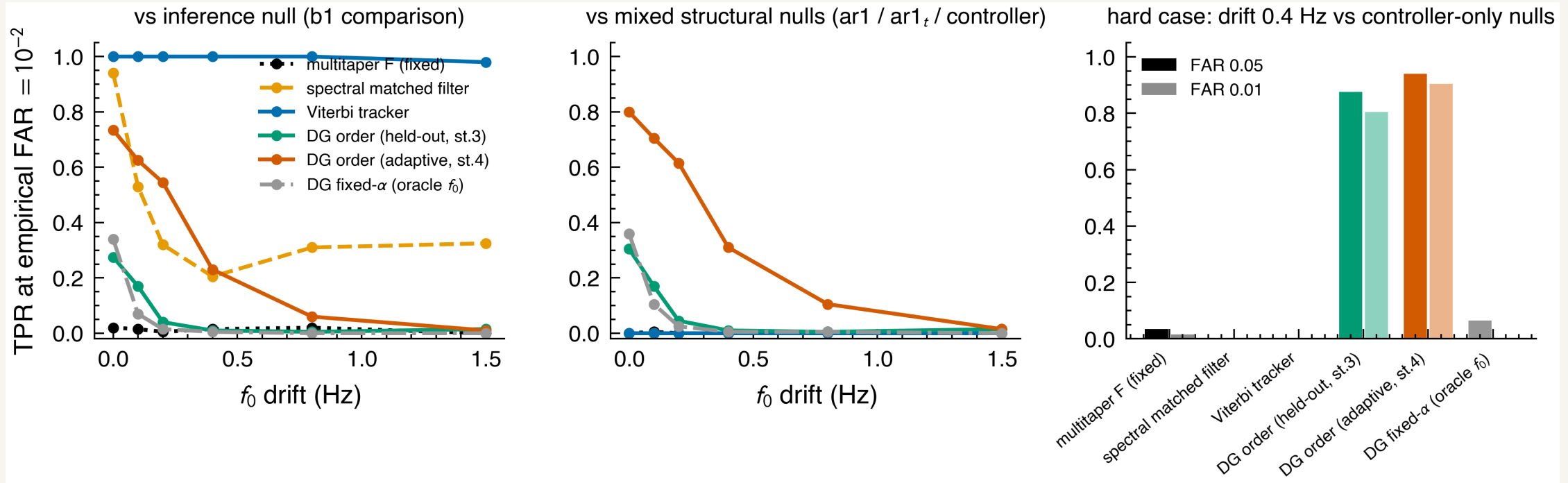
A repeatable physical schedule whose cadence may drift.

Not evidence for

The semantic identity of the program or a model update.

CORPUS-FREE CALIBRATION A single trace can be compared with Fourier surrogates under a restricted linear-stationary structural null.

Finding a line and attributing structure are different jobs



SYNTHETIC BAKE-OFF Detection rate at nominal false-alarm rate 0.05. Viterbi dominates against the inference null (left); order-based statistics reject controller-line confusers (middle and right).

Lesson: the detector that best follows a wandering line need not be the detector that best distinguishes its cause.

There are two different “null” problems

	Question	Where the comparison comes from	What calibration means
Rung 1	Is this more structured than a linear-stationary trace?	Fourier surrogates made from the trace itself	Valid only for that structural null
Rung 2	Is this more training-like than stated inference?	An explicit reference population	Conditional on that population and transfer

Adaptive search changes the score distribution. Calibration must rerun the **entire** search on every surrogate or null trace.

The surrogate method does not make controller cycles part of the null, and the classifier cannot avoid stating what “non-training” means.

Robustness is not binary; it is a cost frontier

How much useful work must the prover give up to make the trace look benign?

Unlimited adversary: passive semantic verification is impossible.

Operational question: which evasions are cheap, and which are costly?

Attack families

idle jitter · real-work variation · drift ·
phase diffusion · harmonic relocation ·
shaping · dilution · meter degradation

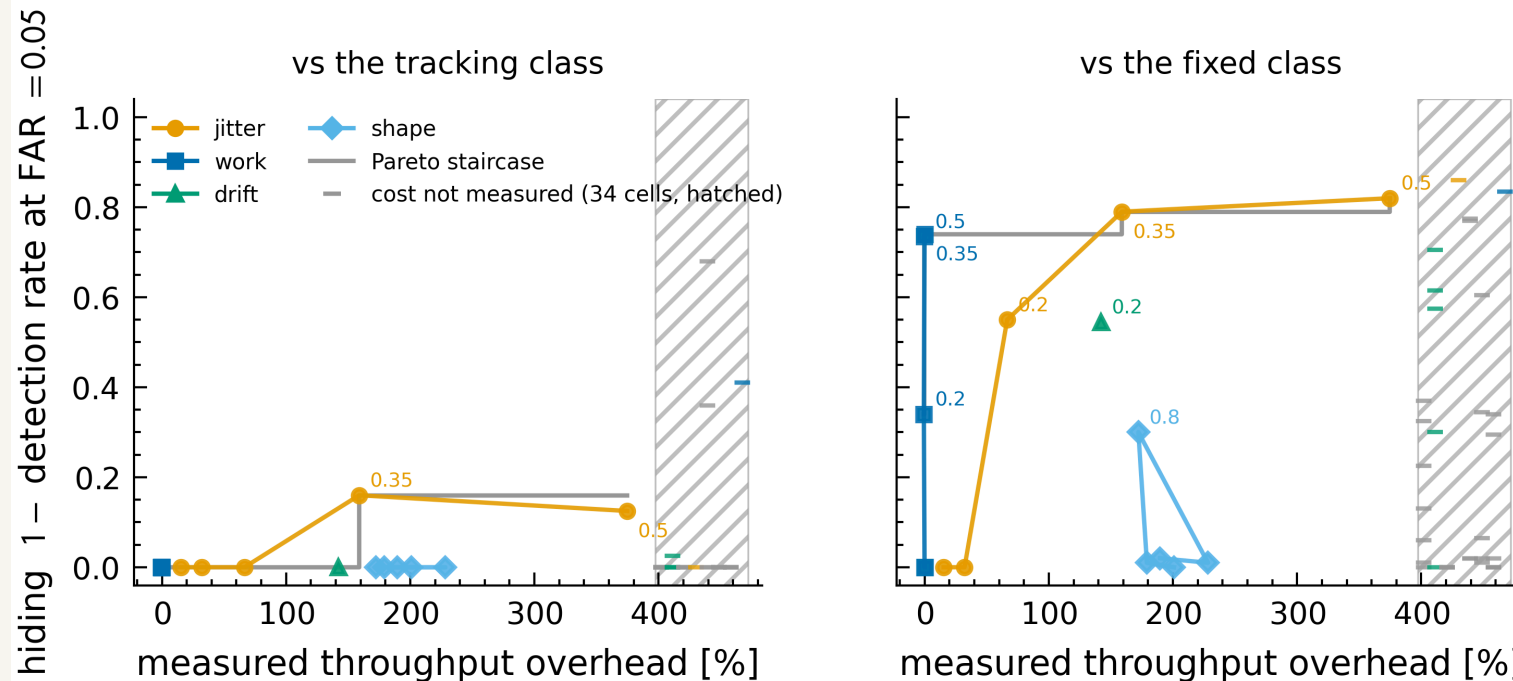
50 SYNTHETIC CELLS

16 MEASURED COST ANCHORS

The attacker knows the detector family; unpriced cells remain visibly unpriced.

The cost frontier is sharply asymmetric

Cost of hiding: what measured adversary cost buys against each detector class



SYNTHETIC DETECTION

MEASURED THROUGHPUT ANCHORS

Fixed-frequency tests lose 0.74 detection power at $\approx 0\%$ overhead; the tracking envelope loses only 0.16 at 159% overhead. Hiding = 1 - best detection rate in class, at nominal false-alarm rate 0.05.

A QUALIFICATION THAT MATTERS OPERATIONALLY

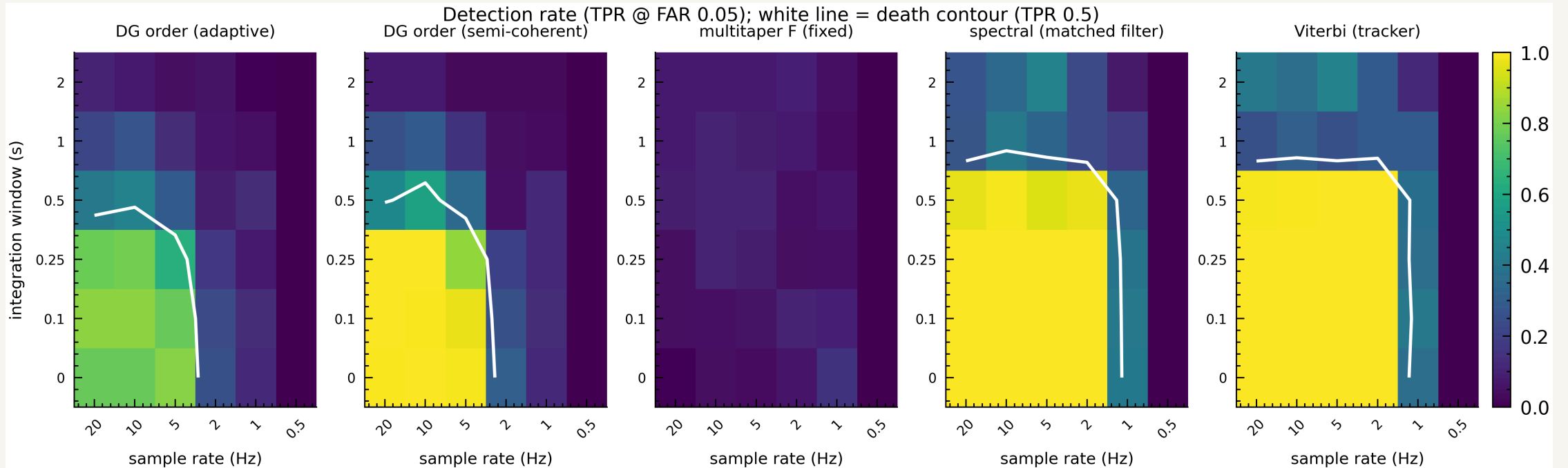
The frontier is a detector-class envelope—not yet one deployed verifier

Method	Follows variable real work	Rejects controller-line confusers	Role in current evidence
Raw Viterbi score	Strong	Weak	Robust path finder
Order-domain tests	Degrade earlier	Strong	Structural attribution

We have not yet demonstrated one jointly calibrated decision rule that inherits both strengths across the full frontier.

A real verifier may run every algorithm it wants—but it must calibrate the final combined decision policy, including selection among them.

The meter is part of the theorem



Sample
at least ≈ 2 Hz

Integrate
no longer than ≈ 0.5 s

Avoid
a deep in-band notch

SCENARIO-SPECIFIC BOUNDARY A 1 s boxcar nulls the 1 Hz band centre; a 1 Hz sampler crosses the cadence-band Nyquist edge.

Against the stated inference population, the classifiers look perfect

1.00

Physics score AUC

1.00

Fitted linear AUC

1.00

Random forest AUC

The result persists across the tested duration, frequency-band, controller, and coloured-noise shifts.

But accuracy against a declared null is not the same thing as semantic validity.

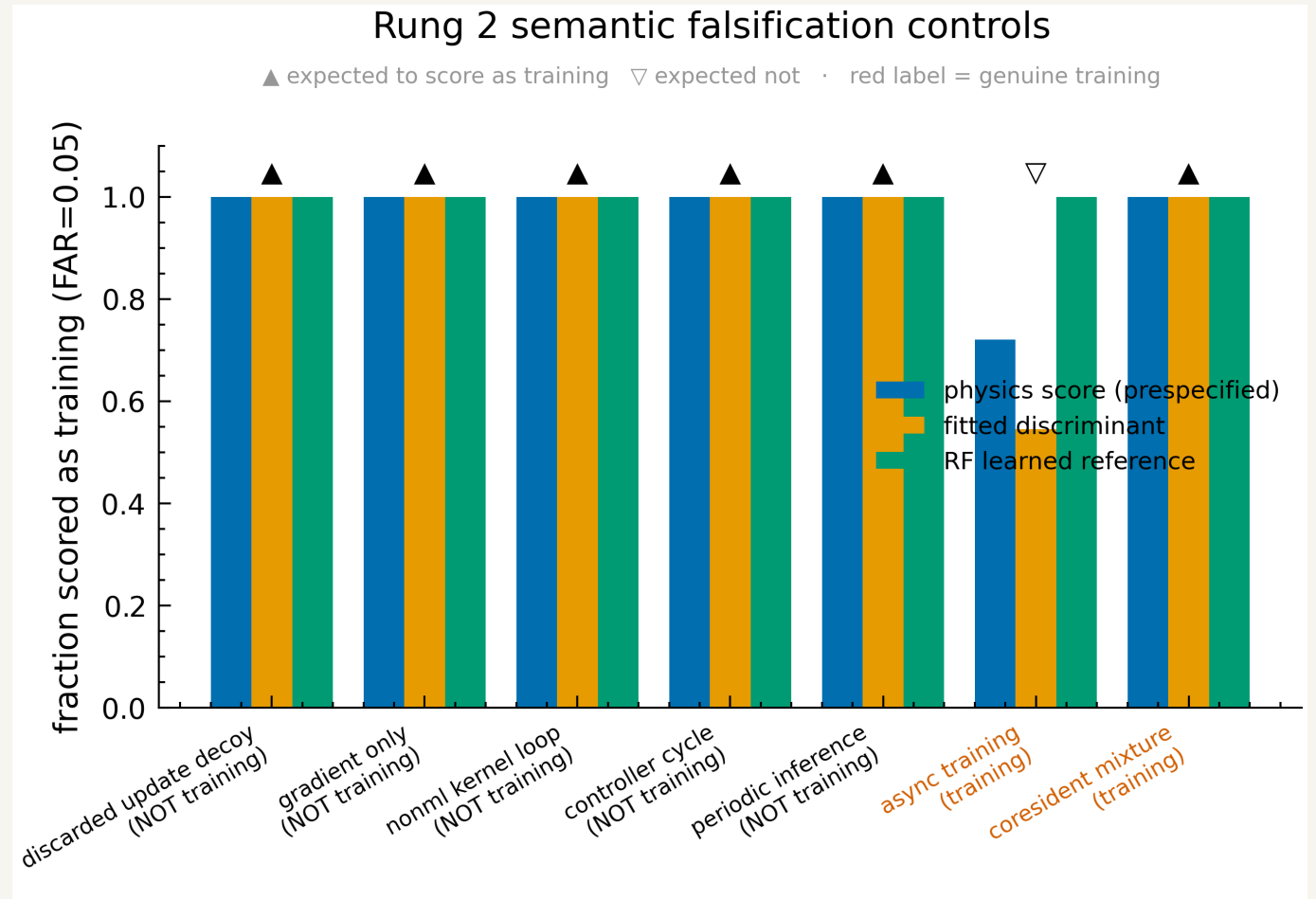
SYNTHETIC POPULATIONS • N = 200 PER CLASS

The semantic controls overturn the naive conclusion

Five non-training workloads are called “training” by every rule.

- discarded update
- gradient only
- non-ML kernel loop
- controller cycle
- periodic inference

The meter certifies a physical schedule, not training semantics.

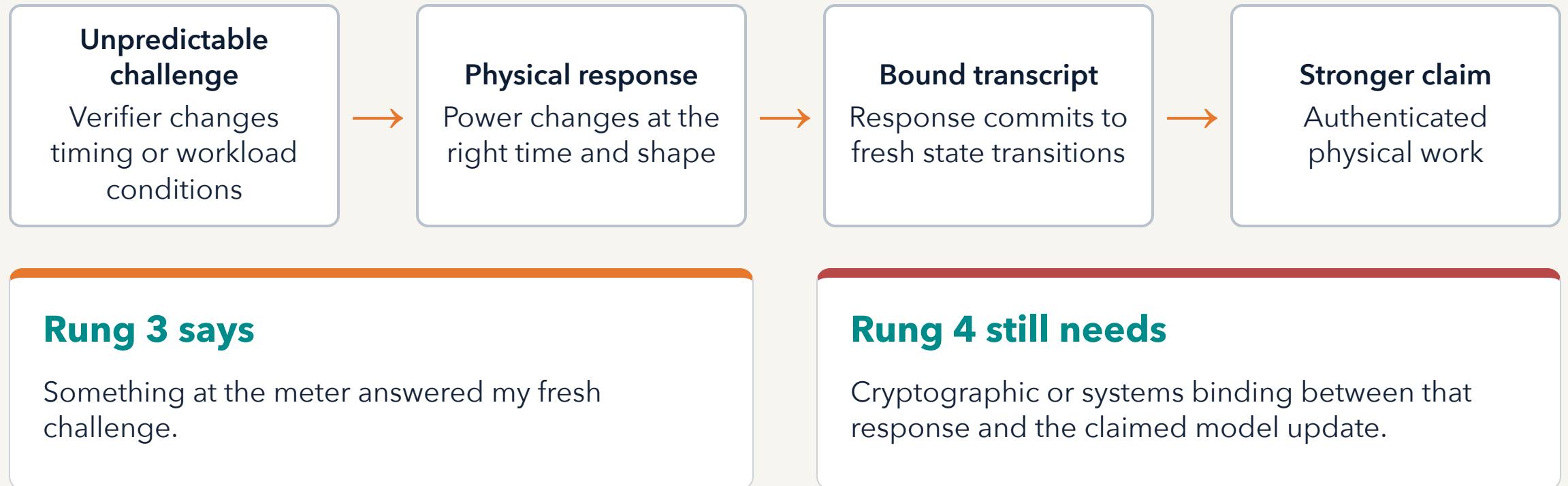


SYNTHETIC FALSIFICATION CONTROLS

Fraction classified as training at false-positive rate 0.05 on the stated null.

RUNGS 3 AND 4

Stronger claims require changing the protocol



PROTOCOL SPECIFIED, NOT RUN A co-located dummy responder remains the key gap without binding.

REALITY CHECK

The one measured transport case fails the naive story

0.4%

iteration-cadence modulation of
≈370 W board power

10x

stronger power-controller line in
the relevant band

75x

gap from the scenario model's
modulation depth

Every positive result in this talk is synthetic or literature-parameterized. The A100 trace is a **negative transport case**, not validation.

Falsifiable next step: synchronized multi-GPU communication should deepen and cohere the cadence at an external rack or wall meter.

What the present work establishes—and what remains open

Established in scope

- A claim ladder with explicit information costs and ceilings
- Corpus-free calibration for a restricted structural null
- A synthetic de-periodicisation frontier with measured cost anchors
- A scenario-specific minimum meter boundary
- A constructive semantic impossibility result

Not yet established

- External-meter detection on a real synchronized cluster
- A joint rule robust to both controller confusers and variable work
- The learning-efficiency cost where the tracker finally fails
- Passive proof that model state was updated
- An implemented active challenge and binding protocol

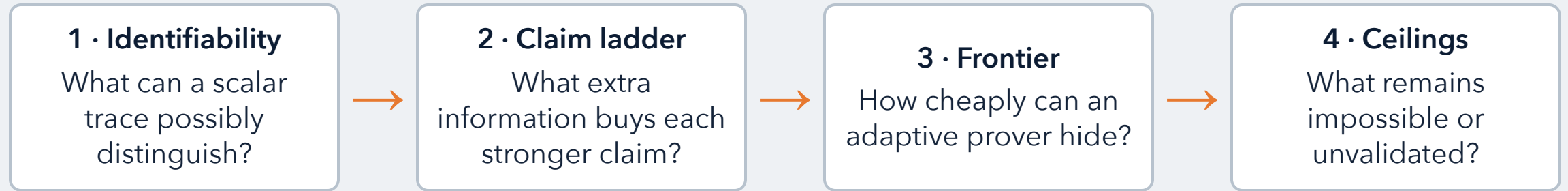
This is a theory-and-methods result: a map of defensible claims, useful tests, attack costs, and hard limits.

Three things to remember

- 1 Passive power can provide calibrated evidence about a **physical schedule**.
 - 2 Against an adversary, the meaningful result is a **detection-utility frontier**, not "robust" or "not robust."
 - 3 Claims about **training semantics** require extra information: a stated null, an active challenge, and ultimately binding to state transitions.
-

The meter is useful precisely when we are exact about what it cannot say.

APPENDIX The paper spine implied by the talk



Detector construction supports the story; it should not become the story.

Method role

Operationalize the weakest claim.

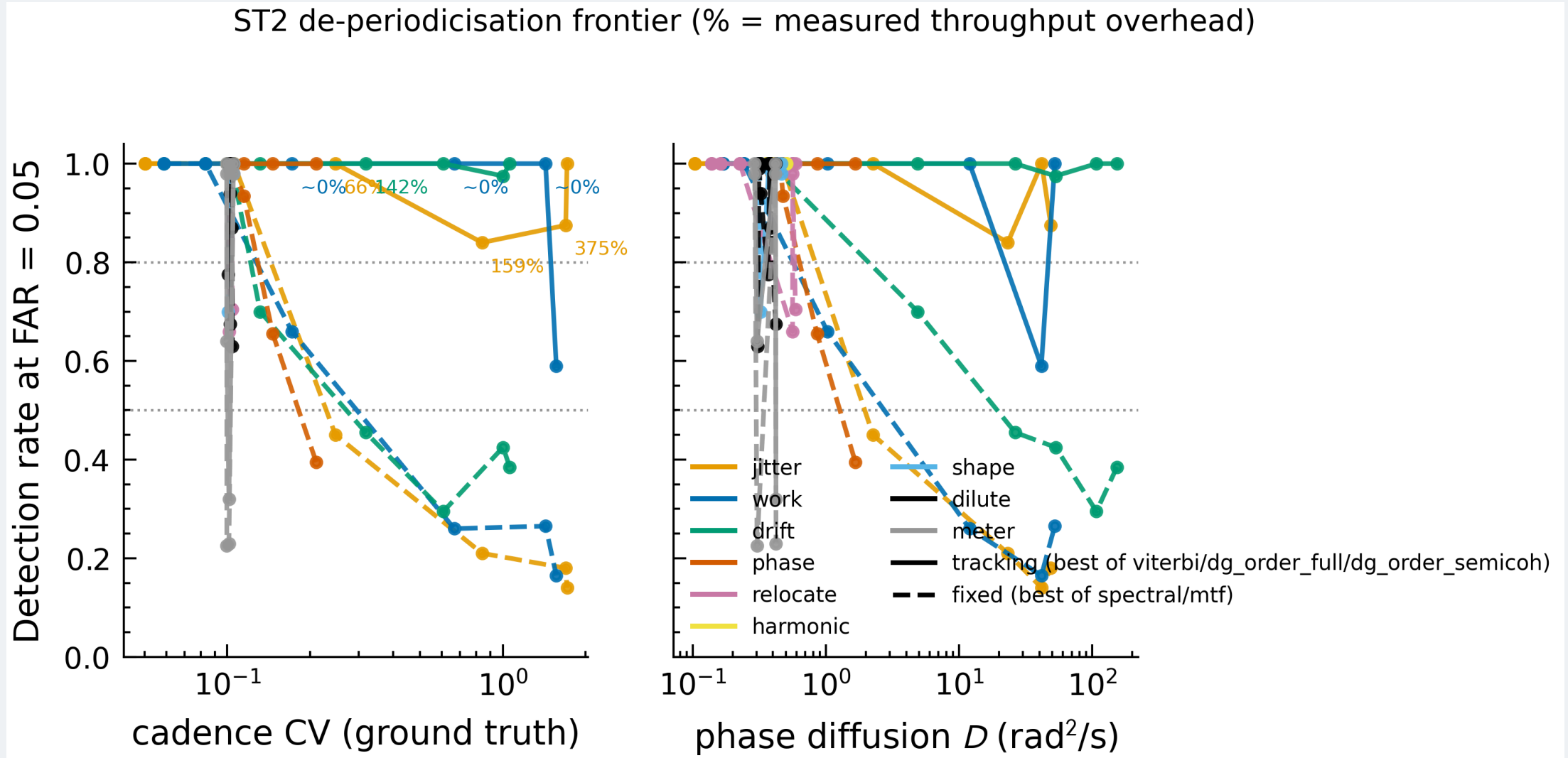
Empirical role

Price adaptive evasion.

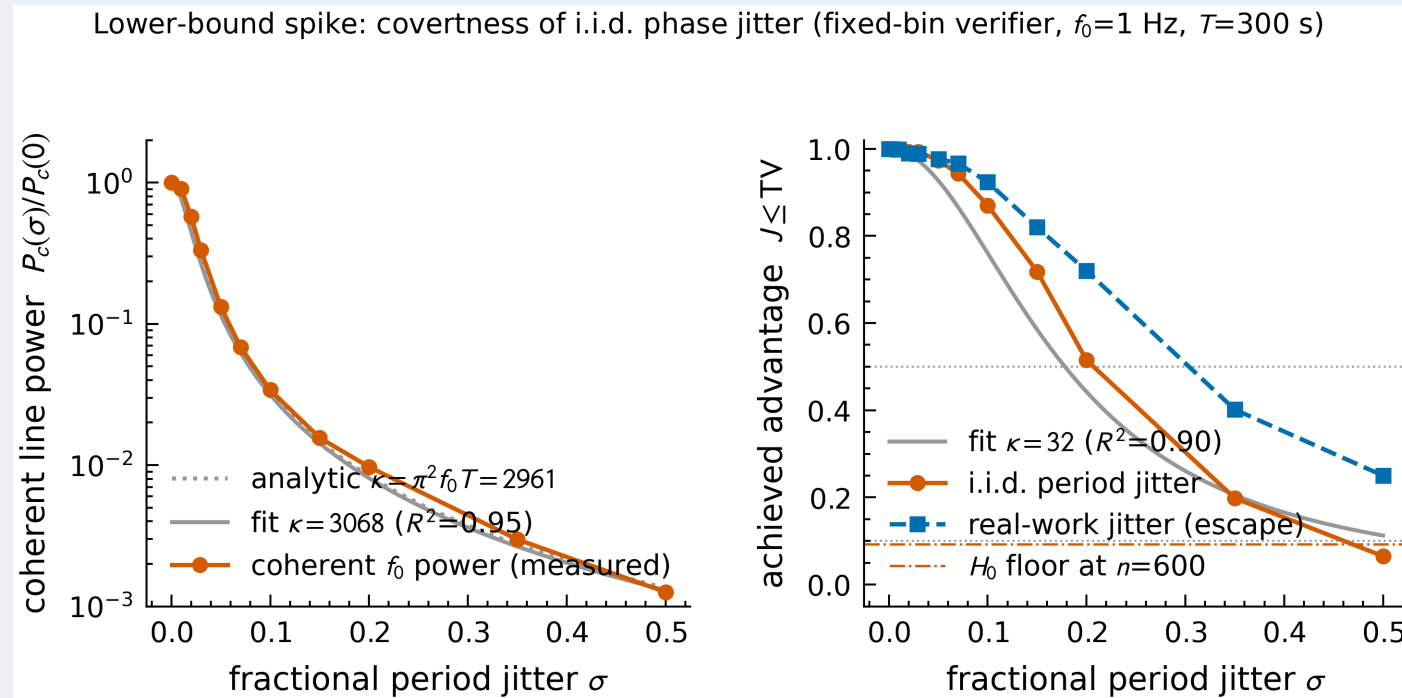
Control role

Falsify semantic overclaiming.

APPENDIX Full de-periodicisation frontier



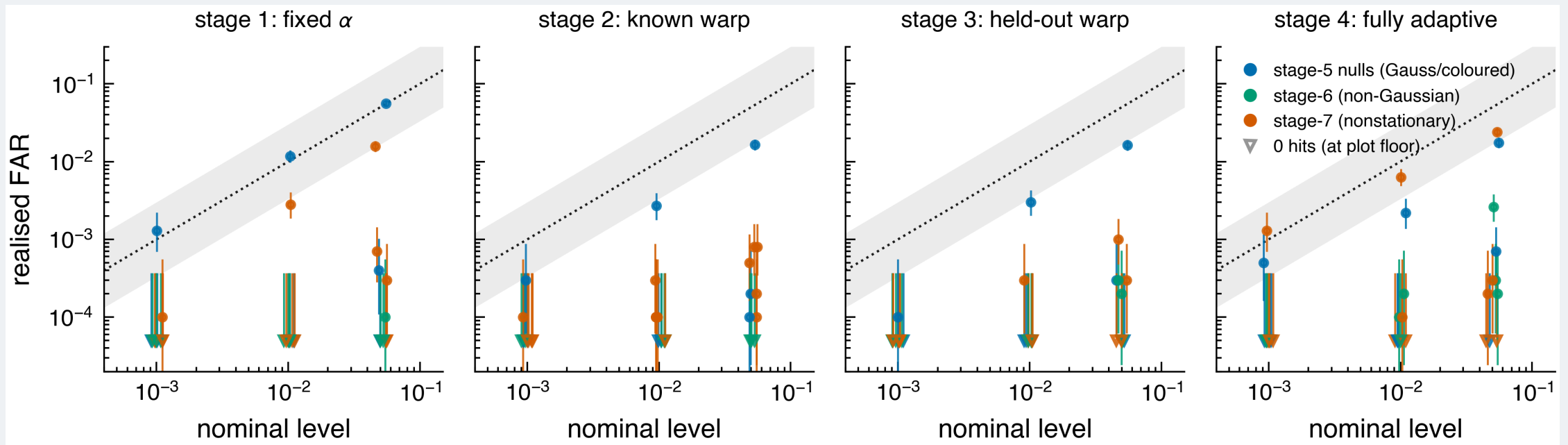
APPENDIX A limited lower-bound direction



If an exhibited test still has advantage J , then the attacked and benign trace distributions are at least J apart in total variation.

The converse does not follow: failure of our detector is not proof of covertness. The cost bridge shown here is specific to i.i.d. timing jitter and a fixed-bin verifier.

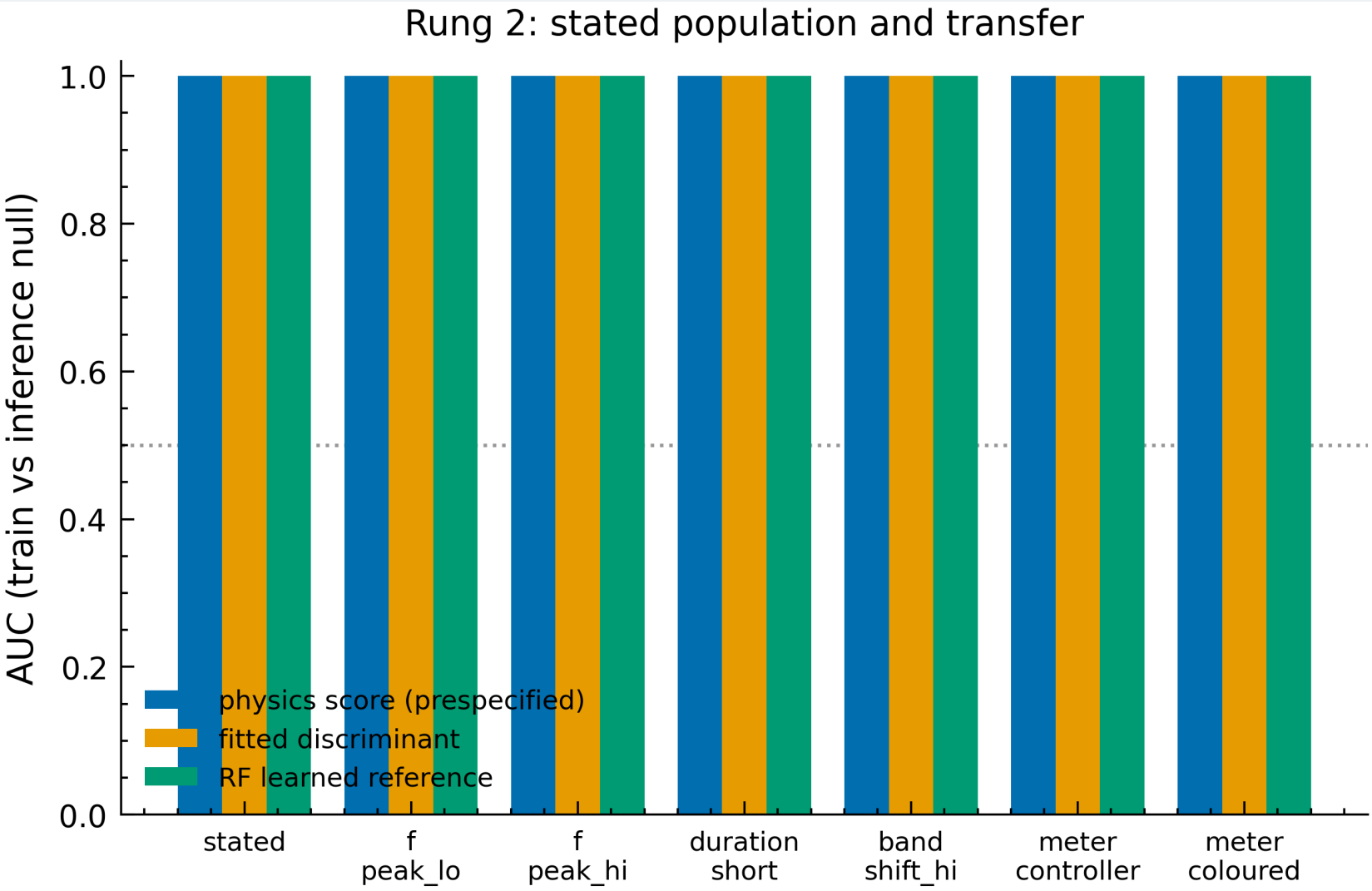
APPENDIX Why analytic calibration was rejected



The adaptive pipeline does not preserve the simple analytic reference law. Per-trace surrogates rerun the complete search; empirical false-alarm control is reported only over the tested structural null family.

APPENDIX

Rung 2 transfer tests



All three rules separate the scenario training population from the stated inference population across the tested shifts. This is conditional discrimination, not semantic identification

APPENDIX Evidence map

Claim	Evidence used here	Main missing validation
Structural detector	Synthetic bake-off + surrogate calibration	External aggregate traces
Adversarial frontier	Synthetic attacks + 16 measured cost anchors	Prices for 34 cells; learning cost
Meter boundary	Scenario observation sweep	Real meter transfer functions
Conditional classifier	Synthetic stated null + transfer shifts	Representative deployment null
Semantic ceiling	Constructive decoys + identifiability argument	Cannot be repaired passively
Active verification	Protocol specification	Implementation and binding